



University of Moratuwa

Department of Electronic & Telecommunication Engineering

EN2150 - Communication Network Engineering

Design of Local Area Network

Backbone Network & ENTC Building Internal LAN

Team Synex

230033J	Amarasinghe A.A.D.K.
230082G	Bandara W.D.A.C.
230164K	Dissanayake R.K.T.
230629R	Tennakoon U.G.R.B.

3rd April 2026

Contents

1	Introduction	3
2	Approach to Backbone Design	3
2.1	Design Objectives	3
2.2	Topology Selection: Distributed Ring with Redundant Shortcut Links . . .	3
2.3	Hierarchical Model	5
2.4	Bandwidth Sizing	5
2.5	Internet Connectivity	5
3	Backbone Network - Building Nodes and Link Bandwidths	5
3.1	Building Nodes	5
3.2	Ring Links and Bandwidths	6
4	IP Addressing Scheme	8
4.1	IPv4 Addressing	8
4.2	IPv6 Addressing	8
4.3	ENTC VLAN Addressing	9
5	ENTC Building Internal LAN Design	9
5.1	Physical Overview	9
5.2	Core Equipments	9
5.3	Floor-by-Floor Device Deployment	10
5.4	VLAN Design	14
5.5	Centralised Wireless Management from CITEs	15
5.6	Structured Cabling (TIA-568-C)	15
6	Justification for Active Components	16
6.1	Cisco 3650-24PS - Distribution and Core Switch	16
6.2	Cisco 2960-24TT - Access Layer Switch	16
6.3	Cisco 3702i - 802.11ac Lightweight Access Point	16
6.4	Cisco WLC-2504 - Wireless LAN Controller	16
6.5	Cisco 2911 ISR - Core Router	16
7	Justification for Passive Components	17
7.1	OS2 Single-Mode Fibre - Backbone Links	17
7.2	OM4 Multi-Mode Fibre - Building Risers	17
7.3	Cat6A UTP - Horizontal Cabling	17
7.4	Patch Panels	17
8	Features and Specifications of Routers and Switches	18
8.1	Cisco 3650-24PS - Layer 3 Distribution/Core Switch	18
8.2	Cisco 2960-24TT - Layer 2 Access Switch	18
8.3	Cisco 2911 ISR - Core Router	18
8.4	Cisco WLC-2504 - Wireless LAN Controller	19
9	Bill of Quantities	19
9.1	Backbone Network - Active Components	19
9.2	Backbone Network - Passive Components	19

9.3	ENTC Internal Network - Active Components	20
9.4	ENTC Internal Network - Passive Components	20
10	Simulation Results	21
10.1	ENTC Network Simulation	21
10.2	Backbone Network Simulation	23
11	Packet Tracer Simulation Files	23
12	Conclusion	23

1. Introduction

This report presents the design of a campus-wide Local Area Network for the University of Moratuwa. The design covers two inter-related components: (a) the university backbone network connecting all major academic buildings, administrative offices, the library, and the data centre (Center for Information Technology Services - CITeS); and (b) the detailed internal network design for the Department of Electronic and Telecommunication Engineering (ENTC) building as a reference implementation.

The backbone network is designed for a 20-25 year operational lifetime, with emphasis on redundancy, scalability, and minimized total cost of cabling. A distributed ring topology using 10 Gbps single-mode fibre interconnects 12 building nodes through redundant distribution switches at each node and two redundant core switches at CITeS. Both IPv4 and IPv6 addressing are provisioned with three times the current estimated demand to accommodate future growth.

The ENTC internal network implements a hierarchical two-tier LAN with a Layer 3 core switch providing inter-VLAN routing across three functional VLANs, 12 floor-level Layer 2 access switches, and 14 Lightweight Wi-Fi access points managed centrally from CITeS via a Wireless LAN Controller (WLC).

Both networks are simulated in Cisco Packet Tracer. The logical diagrams produced by the simulation are referenced throughout this report.

2. Approach to Backbone Design

2.1 Design Objectives

The backbone must satisfy the following requirements over its 20-25 year lifespan:

- High availability through redundant physical links and device pairs
- Sufficient bandwidth for simultaneous campus-wide usage (target: ≥ 1 Mbps per device)
- Scalability for new buildings and increased device counts without re-cabling
- Minimized total cost of ownership: upfront cabling cost plus maintenance over the lifetime
- Broadcast containment: each building is a separate IP subnet; broadcasts do not traverse the backbone
- Dual-stack IPv4 and IPv6 support throughout

2.2 Topology Selection: Distributed Ring with Redundant Shortcut Links

A **distributed ring backbone** topology is selected. All 12 building-level distribution nodes are connected in a ring using 10 Gbps single-mode fibre links. Additional shortcut links are added at high-traffic or geographically central node pairs - specifically NOC-ENTC, NOC-ADMIN, ENTC-Sumanadasa, ENTC-IT Faculty, and ENTC-Architecture - to reduce hop count and provide more than two diverse paths between the most heavily used nodes. These are visible as the red links in the backbone Packet Tracer simulation.

Why not a star topology?

A star (all buildings connecting directly to CITEs) creates a single point of failure at the CITEs aggregation point and requires the longest total cable run, since every building needs a direct fibre path to the campus centre. The ring provides at least two independent paths between any node pair at significantly lower total fibre length.

Why not a full mesh?

A full mesh provides maximum path diversity but the number of links grows as $n(n-1)/2$. For 12 nodes this means 66 links - prohibitive in cabling cost and port consumption. The ring plus targeted shortcut links achieves near-mesh resilience for the most critical node pairs at a fraction of the cost.

Why a distributed backbone and not a collapsed backbone?

A collapsed backbone concentrates routing at a single device pair. For a large geographically spread campus this creates a scalability bottleneck and limits horizontal expansion. The distributed model places a Layer 3 switch at each building, segmenting the campus into independent routable subnets, distributing processing load, and allowing each building to continue local operation even if the backbone has a partial outage.

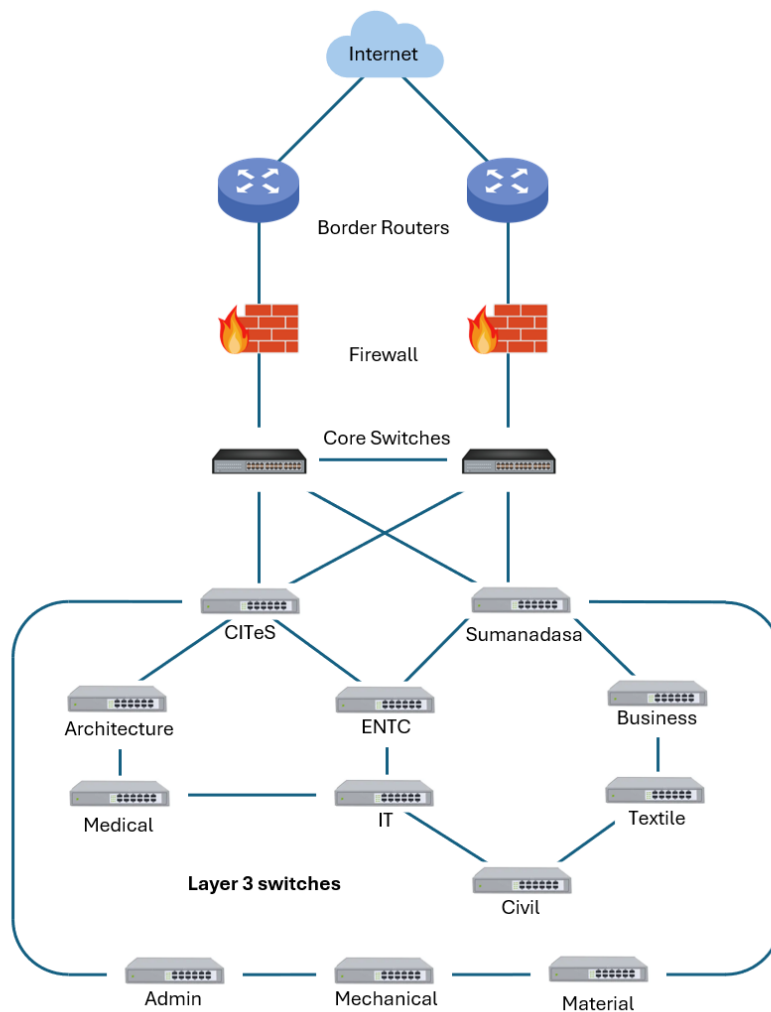


Figure 1: Logical Architecture of the Backbone Network

2.3 Hierarchical Model

The backbone follows the standard three-layer hierarchical model:

Core Layer

Two redundant Cisco 3650-24PS Layer 3 switches at CITeS (NOC) form the root of the network. They handle inter-node routing for the entire campus and provide the uplink to the internet via the border routers and firewalls. Two units are deployed in an active/standby OSPF arrangement.

Distribution Layer

One Cisco 3650-24PS Layer 3 switch per building node. Each distribution switch connects to the backbone ring via 10 Gbps SM fibre uplinks and connects downstream to the building's internal access switches. Layer 3 at the distribution layer means inter-VLAN routing is performed locally at each building, keeping intra-building traffic off the backbone.

Access Layer

Cisco 2960-24TT Layer 2 switches inside each building, one per floor zone or major laboratory. These connect end devices and PoE wireless access points to the distribution switch via 1 Gbps links.

2.4 Bandwidth Sizing

A minimum of 1 Mbps per connected device is used as the design baseline. The university has fewer than 10,000 users; a 10 Gbps backbone link supports 10,000 simultaneous 1 Mbps flows. Crucially, the OS2 single-mode fibre cable plant installed today supports transceiver-only upgrades to 40 Gbps or 100 Gbps, meaning the physical medium does not need replacement even if bandwidth requirements grow significantly over the 20–25 year life.

2.5 Internet Connectivity

Dual border routers (Cisco 2911 ISR) connect to the LEARN network via the CITeS NOC. A redundant pair of next-generation firewalls (Cisco Firepower 2110) sits between the border routers and the core switches, providing stateful inspection, NAT, VPN termination, and intrusion prevention. The dual-router, dual-firewall HA arrangement eliminates single points of failure on the internet path.

3. Backbone Network - Building Nodes and Link Bandwidths

3.1 Building Nodes

Table 1 lists all backbone distribution nodes. Each node is served by a Cisco 3650-24PS Layer 3 distribution switch. Building-level 2960-24TT access switches connect downstream from each distribution switch.

Table 1: Backbone Distribution Nodes

	Node / Switch Name	Campus Location	Connected 2960-24TT Access Switches
1	NOC (CITeS)	CITeS Data Centre	Library, Integrated Design, Open Distance Training, Rubert Peries Hall, Building Economics
2	ENTC	ENTC Building	See Section 5
3	Architecture Dept.	Faculty of Architecture	New Exam Hall, NA1/2
4	Medical Faculty	Faculty of Medicine	–
5	IT Faculty	IT Building	UNESCO Center, Female Hostel, FOT Phase 2, New Classroom, Research Center
6	CIVIL Department	Civil Eng. Building	Hostal A
7	Textile Department	Textile Faculty	Fashion Design, TML Building
8	Business Faculty	Business Faculty	Decision Science, Math Department, L Canteen
9	Material Department	Materials Eng. Building	CSE Dept., Chemical Eng. Dept., OLD Gym, Institute of Technology
10	Sumanadasa Building	Sumanadasa Block	Dean's Office, EE Department, Dept. of Building Economics, Steel Building
11	ADMIN	Administration Complex	Post Graduate Division, Exam Division
12	Mechanical Complex	Mechanical Eng. Building	New Conference Hall, Enterprise Building, Maritime Studies Building, Mechanical Department

3.2 Ring Links and Bandwidths

All primary backbone links carry **10 Gbps** over **OS2 single-mode fibre**. The ring connectivity is:

- NOC ↔ Architecture Department
- Architecture Department ↔ ENTC
- ENTC ↔ IT Faculty
- IT Faculty ↔ CIVIL Department
- CIVIL Department ↔ Textile Department
- Textile Department ↔ Business Faculty
- Business Faculty ↔ Material Department
- Material Department ↔ Sumanadasa Building

- Sumanadasa Building ↔ ADMIN
- ADMIN ↔ NOC
- ADMIN ↔ Mechanical Complex
- Mechanical Complex ↔ Material Department

Additional redundant shortcut links (also 10 Gbps SM fibre):

- NOC ↔ ENTC
- NOC ↔ ADMIN
- ENTC ↔ Sumanadasa Building

Building access links: 1 Gbps copper or OM4 multi-mode fibre from each distribution switch to building-level 2960-24TT access switches.

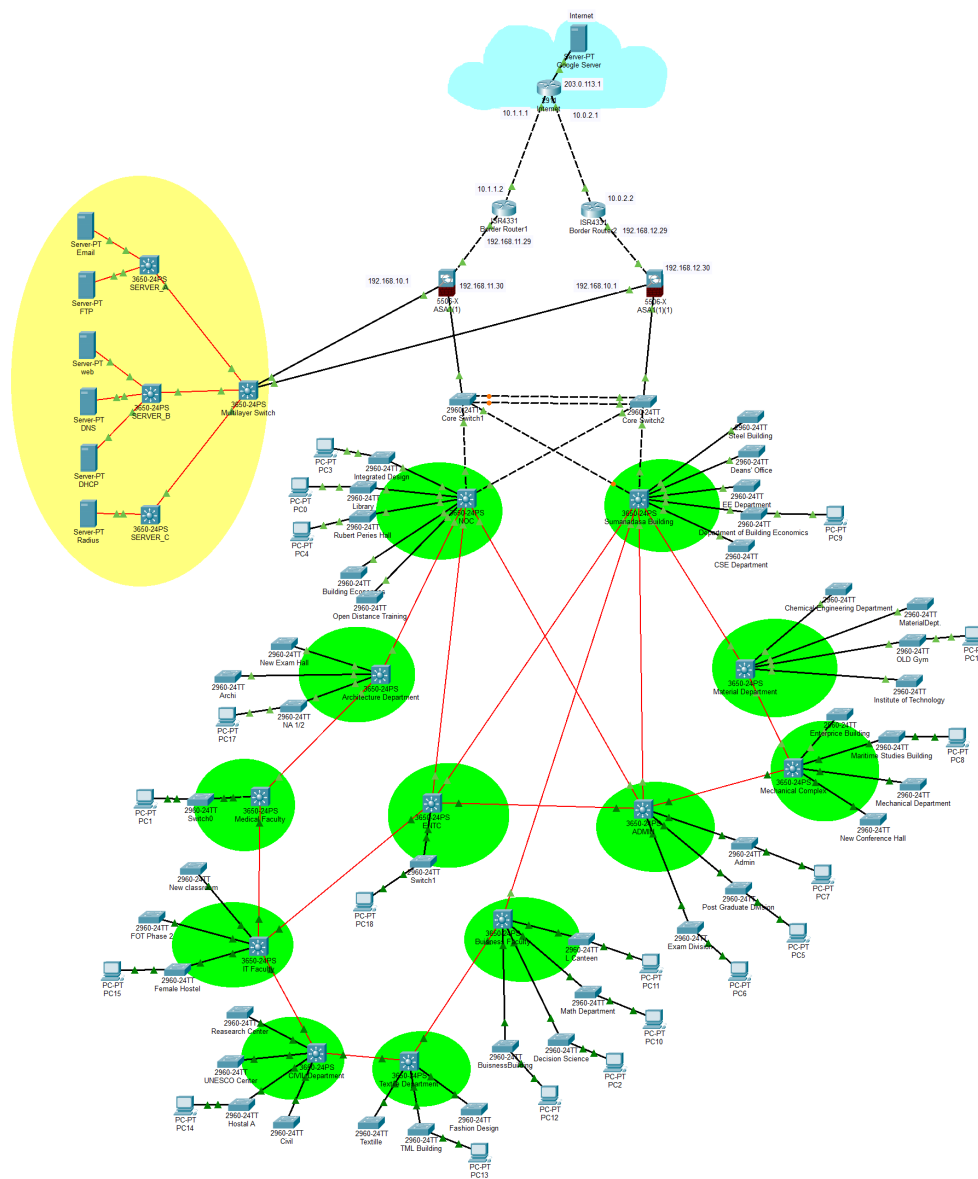


Figure 2: Backbone Network Logical Topology

4. IP Addressing Scheme

4.1 IPv4 Addressing

A Class B private address block (172.16.0.0/16) is used for the entire campus. Each backbone node is allocated a subnet sized according to three times the current user population to accommodate long-term growth. Subnets are power-of-two aligned.

Table 2: IPv4 Backbone Addressing

Node	Needed Hosts	Prefix	Network / Mask	Usable IPs
Sumanadasa	8192	/19	172.16.0.0/19	8190
Civil	8192	/19	172.16.32.0/19	8190
CITeS	4096	/20	172.16.64.0/20	4094
Architecture	4096	/20	172.16.80.0/20	4094
IT	8192	/19	172.16.96.0/19	8190
ENTC	2048	/21	172.16.112.0/21	2046
Mechanical	2048	/21	172.16.120.0/21	2046
Textile	2048	/21	172.16.128.0/21	2046
Medical	2048	/21	172.16.136.0/21	2046
Admin	512	/23	172.16.144.0/23	510
Business	1024	/22	172.16.146.0/22	1022

4.2 IPv6 Addressing

The university is allocated the prefix 2400:ee00:1000::/48. Variable-length IPv6 subnetting is applied according to building size.

Table 3: IPv6 Backbone Addressing

Node	IPv6 Prefix	Prefix Length
Sumanadasa	2400:ee00:1000:0000::	/51
Civil	2400:ee00:1000:2000::	/51
CITeS	2400:ee00:1000:4000::	/52
Architecture	2400:ee00:1000:5000::	/52
IT	2400:ee00:1000:6000::	/51
ENTC	2400:ee00:1000:7000::	/53
Mechanical	2400:ee00:1000:7800::	/53
Textile	2400:ee00:1000:8000::	/53
Medical	2400:ee00:1000:8800::	/53
Admin	2400:ee00:1000:9000::	/54
Business	2400:ee00:1000:9400::	/54

4.3 ENTC VLAN Addressing

VLAN ID	Name	Purpose	IPv4 Subnet	Gateway
10	STUDENTS	ENTC students	172.16.112.0/22	172.16.112.1
20	STAFF	ENTC staff	172.16.116.0/23	172.16.116.1
30	LAB	ENTC lab computers	172.16.118.0/23	172.16.118.1

5. ENTC Building Internal LAN Design

5.1 Physical Overview

The ENTC building has seven occupied levels: Ground Floor, Lower Ground Floor, 0.5 Floor, 1st Floor, 2nd Floor, 3rd Floor, and 3.5 Floor. The network room housing the ENTC Core Switch and associated equipment is on the Ground Floor. All floor-level switches connect back to the Core Switch via fibre or copper uplinks routed through the building's riser shaft.

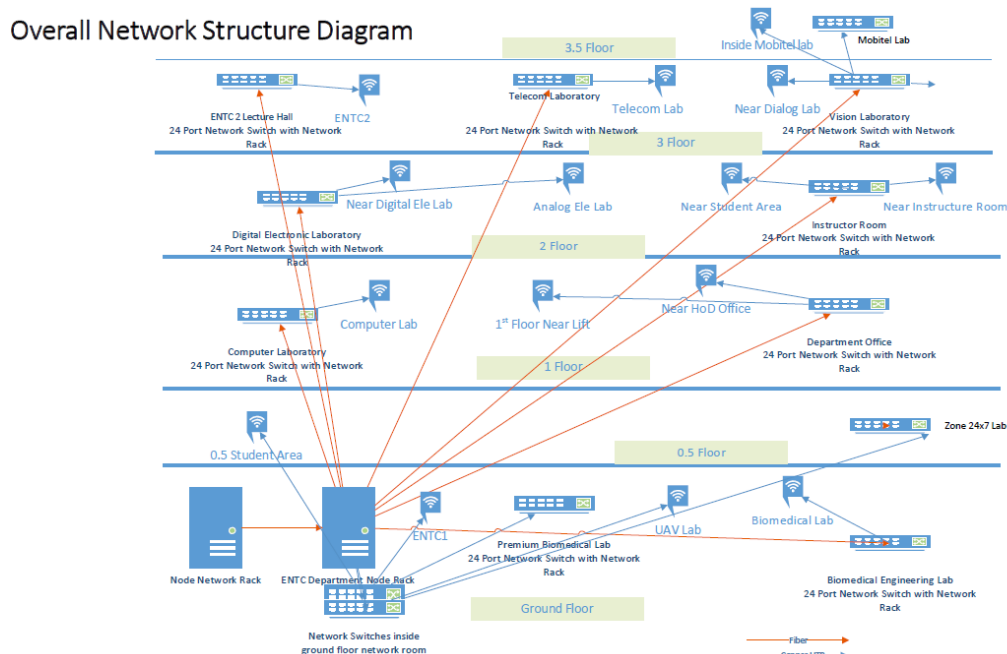


Figure 3: ENTC Network Structure (adapted from presentation slides by Mr. Damith Kandage)

5.2 Core Equipments

The ENTC core switch, a Cisco 3650-24PS Layer 3 switch located in the Ground Floor Network Room, functions as the local aggregation point for the building network. It performs inter-VLAN routing for ENTC VLANs, aggregates all access layer switches within the building, and provides the uplink towards the university backbone. This switch operates as the default gateway for ENTC subnets and participates in the campus dynamic routing protocol with the backbone.

All higher-level control components, including the backbone gateway router and centralized

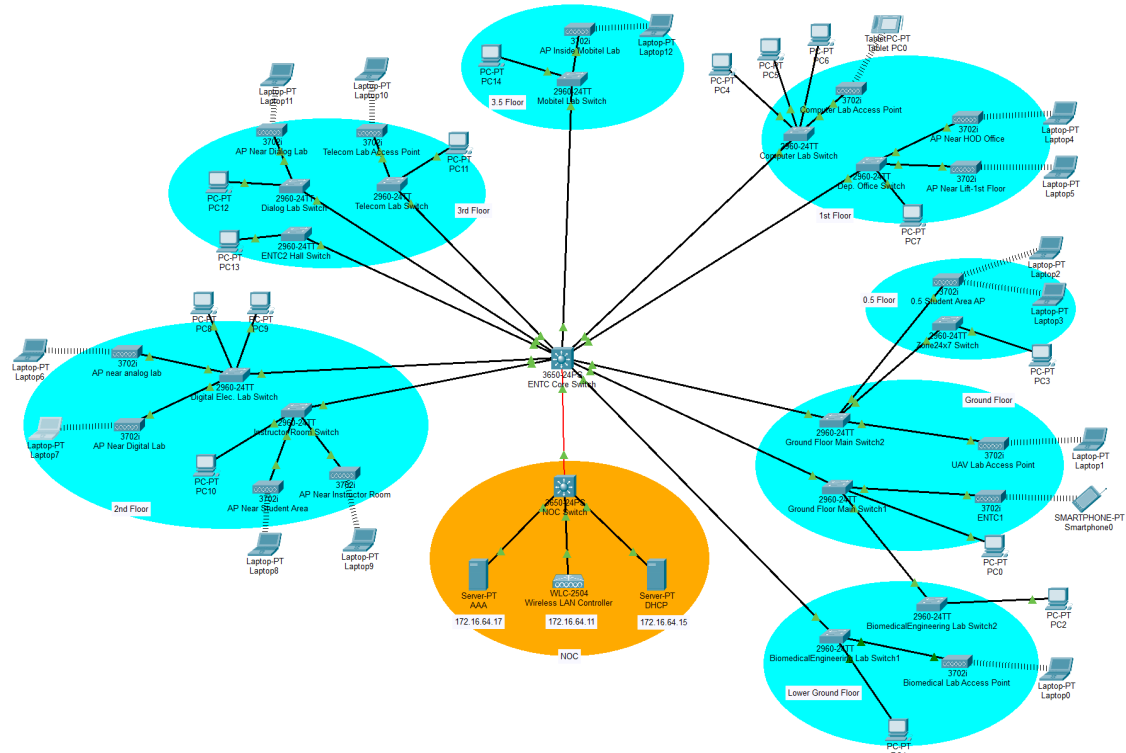


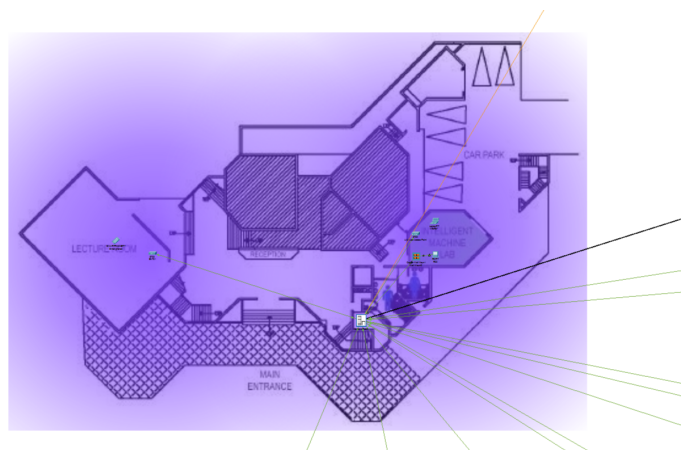
Figure 4: ENTC Network Logical Topology

wireless management infrastructure, are located at the CITEs premises. The ENTC core switch connects to CITEs through a single-mode fibre uplink, where routing decisions, Internet connectivity, and centralized network services are handled. This design follows the university's centralized network management approach, reducing equipment redundancy within individual buildings while simplifying maintenance and long-term scalability.

5.3 Floor-by-Floor Device Deployment

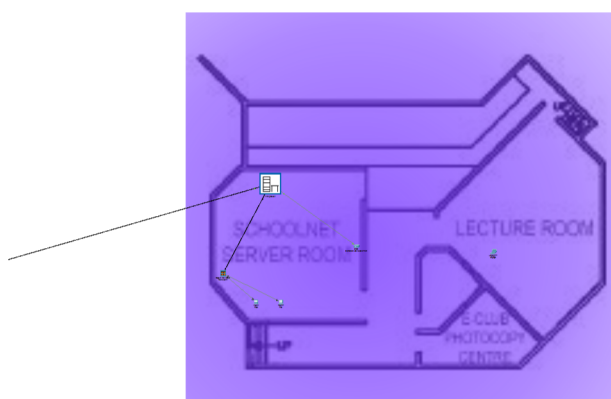
Ground Floor

Device	Model	Connectivity
Ground Floor Main Switch1	2960-24TT	Uplink to Core Switch
Ground Floor Main Switch2	2960-24TT	Uplink to Core Switch
ENTC1 Access Point	3702i	Main Switch1 → WLC
UAV Lab AP	3702i	BioEng Lab Switch2 → WLC

**Figure 5:** Ground Floor Physical Topology

Lower Ground Floor

Device	Model	Connectivity
BiomedEng. Lab Switch1	2960-24TT	Uplink to Core Switch
BiomedEng. Lab Switch2	2960-24TT	Uplink to Core Switch
Biomedical Lab AP	3702i	BioEng Lab Switch1 → WLC

**Figure 6:** Lower Ground Floor Physical Topology

0.5 Floor

Device	Model	Connectivity
Zone24x7 Switch	2960-24TT	Uplink to Core Switch
0.5 Student Area AP	3702i	Zone24x7 Switch → WLC

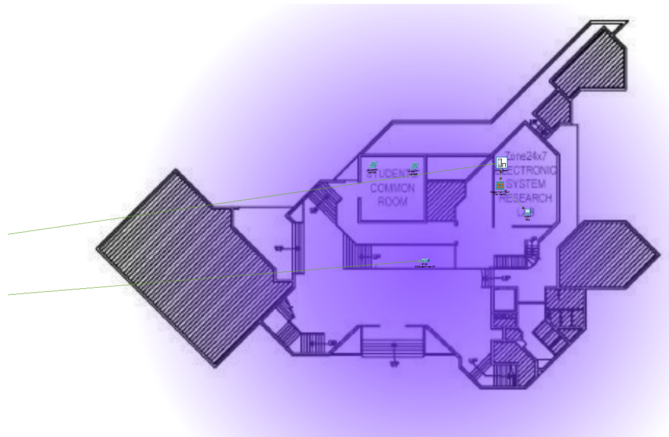


Figure 7: 0.5 Floor Physical Topology

1st Floor

Device	Model	Connectivity
Computer Lab Switch	2960-24TT	Uplink to Core Switch
Dep. Office Switch	2960-24TT	Uplink to Core Switch
Computer Lab AP	3702i	Computer Lab Switch → WLC
AP Near HoD Office	3702i	Dep. Office Switch → WLC
AP Near Lift – 1st Floor	3702i	Dep. Office Switch → WLC

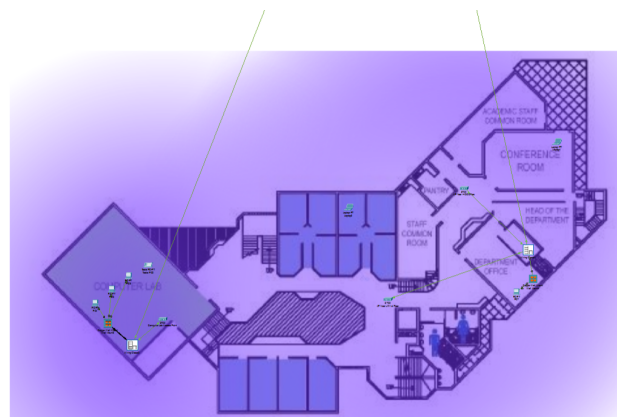
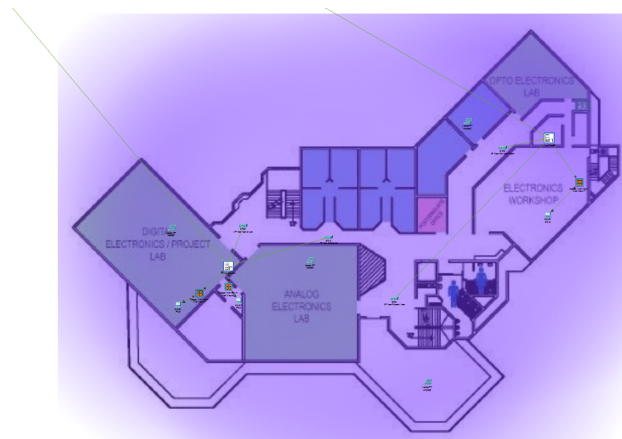


Figure 8: 1st Floor Physical Topology

2nd Floor

Device	Model	Connectivity
Digital Elec. Lab Switch	2960-24TT	Uplink to Core Switch
Instructor Room Switch	2960-24TT	Uplink to Core Switch
AP Near Digital Lab	3702i	Digital Lab Switch → WLC
AP Near Analog Lab	3702i	Digital Lab Switch → WLC
AP Near Instructor Room	3702i	Instructor Room Switch → WLC
AP Near Student Area	3702i	Instructor Room Switch → WLC

**Figure 9:** 2nd Floor Physical Topology**3rd Floor**

Device	Model	Connectivity
ENTC2 Hall Switch	2960-24TT	Uplink to Core Switch
Dialog Lab Switch	2960-24TT	Uplink to Core Switch
Telecom Lab Switch	2960-24TT	Uplink to Core Switch
AP Near Dialog Lab	3702i	Dialog Lab Switch → WLC
Telecom Lab AP	3702i	Telecom Lab Switch → WLC

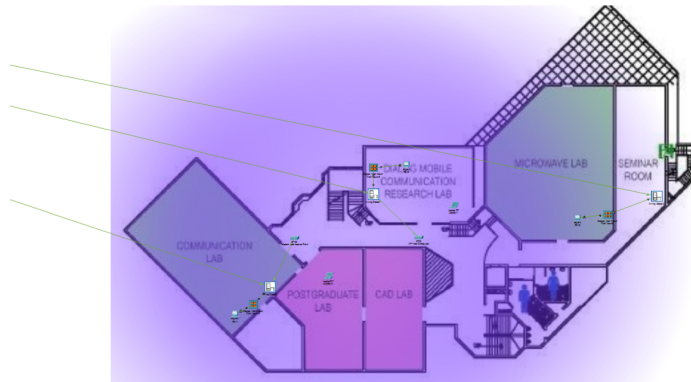


Figure 10: 3rd Floor Physical Topology

3.5 Floor

Device	Model	Connectivity
Mobitel Lab Switch	2960-24TT	Uplink to Core Switch
AP Inside Mobitel Lab	3702i	Mobitel Lab Switch → WLC



Figure 11: 3.5 Floor Physical Topology

5.4 VLAN Design

Rather than per-floor VLANs, the ENTIC network is segmented into three functional VLANs that span the entire building. 802.1Q trunks carry all three VLANs from the Core Switch to every floor access switch. The Core Switch (3650-24PS) SVIs perform inter-VLAN routing. DHCP scopes are served by Server at CITeS with the Core Switch SVIs configured as DHCP relay agents.

VLAN Name	ID	IPv4 Subnet	Description
Students	10	172.16.112.128/22	Student wireless; SSID student WPA2-Enterprise
Staff	20	172.16.116.0/23	Staff wired desks; WPA2-Enterprise SSID Staff
Labs	30	172.16.118.0/23	Lab computers and lab APs; SSID Labs

5.5 Centralised Wireless Management from CITEs

All 14 Cisco 3702i APs operate in Lightweight (CAPWAP) mode. They hold no local configuration; all SSID, security, VLAN mapping, and RF policies are pushed from the Cisco WLC-2504 at Network Operating Center (NOC), which is itself administered centrally from CITEs. This means SSID policy changes, password rotation, new SSID provisioning, and RF channel optimisation are made once at CITEs and propagate to all APs across all buildings simultaneously.

SSID	Security	VLAN	Authentication
Student wifi (student)	WPA2-PSK	VLAN 10	Pre-shared key
Staff wifi (Staff)	WPA2-Enterprise	VLAN 20	RADIUS server at CITEs
Lab wifi (Lab)	WPA2-Enterprise	VLAN 30	RADIUS server at CITEs

5.6 Structured Cabling (TIA-568-C)

Backbone / Riser Cabling (Vertical)

OM4 50/125 μ m multi-mode fibre runs vertically through a dedicated fire-rated riser shaft, connecting each floor's Telecommunications Enclosure (TE) to the Main Distribution Frame (MDF) on the Ground Floor. OM4 supports 10 Gbps over 400 m, comfortably covering all floor-to-floor distances. LC/UPC duplex connectors are used throughout; 12-port LC fibre patch panels terminate fibre at both ends.

Horizontal Cabling

Cat6A UTP (23 AWG, 4-pair) runs from each floor's patch panel to individual RJ45 keystone jack wall outlets. Cat6A is rated for 10GBASE-T over 100 m, future-proofing horizontal infrastructure for the full design life without re-cabling. Each AP mounting point has a dedicated Cat6A run to a PoE+ switch port, providing power and data over a single cable.

Work Area

Each workstation location has a minimum of two RJ45 wall outlets. A 2 m Cat6A patch cord connects the outlet to the end device; a 1 m patch cord connects the patch panel to the switch port at the TE rack.

Patch Panels

24-port Cat6A patch panels terminate horizontal copper runs at each TE and the MDF. 12-port LC duplex fibre patch panels terminate fibre riser runs. Patch panels provide a labelled management point - moves, adds, and changes require only patch cord swaps.

6. Justification for Active Components

6.1 Cisco 3650-24PS - Distribution and Core Switch

The 3650-24PS is a 24-port Layer 3 PoE+ switch with $4 \times 1\text{G}$ SFP uplinks (field upgradeable to 10G SFP+). Selected for both the ENTC core switch and all backbone distribution nodes because:

- Hardware-based inter-VLAN routing at wire speed; no software routing latency
- Native OSPF support enabling automatic ring failover
- PoE+ on all 24 ports powers APs and IP phones without separate injectors
- 24 ports accommodates all building access switches with room for future additions
- Single model across all nodes simplifies spares management over a 20-25 year life

6.2 Cisco 2960-24TT - Access Layer Switch

- 24 ports accommodates all end devices and APs per floor zone
- Layer 2 only is correct for the access layer; routing is handled by the distribution switch
- Lower unit cost than Layer 3 switches; appropriate where routing is not required
- Two 1G uplinks support LACP bonding to the distribution switch for added bandwidth and redundancy

6.3 Cisco 3702i - 802.11ac Lightweight Access Point

- Dual-band (2.4 GHz + 5 GHz), 802.11ac Wave 1, 3×4 MIMO, 1.3 Gbps aggregate radio rate
- Lightweight (CAPWAP) mode: no local configuration; all policies pushed from WLC at CITEs
- Centralized management ensures campus-wide SSID/security policy consistency
- All wireless client traffic is tunnelled to the WLC, inspected, and VLAN-tagged before reaching the wired network

6.4 Cisco WLC-2504 - Wireless LAN Controller

- Supports up to 75 APs and 1000 wireless clients
- Co-located in ENTC network room; centrally administered from CITEs for all campus buildings
- Provides rogue AP detection, Layer 2/3 fast roaming, and RADIUS integration for 802.1X

6.5 Cisco 2911 ISR - Core Router

- Provides ENTC-backbone WAN uplink and handles OSPF peering with the NOC

- Supports IPsec VPN, ACL enforcement, and NAT at the building perimeter
- Expandable with EHWIC-1GE-SFP-CU module for SM fibre uplink to NOC

7. Justification for Passive Components

7.1 OS2 Single-Mode Fibre - Backbone Links

OS2 single-mode fibre (9/125 μm) is used for all inter-building backbone links.

- Supports 10 Gbps (and future 40 Gbps/100 Gbps) over distances up to 10 km without amplification. Campus building separations of 100–500 m are well within this limit.
- Attenuation ≈ 0.35 dB/km at 1310 nm, versus ≈ 3.5 dB/km for OM4 at 850 nm. Lower attenuation provides greater link margin for splices and connectors.
- **Key total-cost-of-ownership argument:** the fibre cable plant is the most expensive and disruptive item to replace. OS2 SM fibre installed today supports at least three transceiver generations (10G $\rightarrow$ 40G $\rightarrow$ 100G) over the design life by replacing only the SFP modules, not the cable.
- HDPE conduit protection for underground runs provides mechanical protection and allows future cable replacement without excavation.

7.2 OM4 Multi-Mode Fibre - Building Risers

OM4 50/125 μm fibre is used for intra-building vertical riser cabling.

- Intra-building distances are well within the 400 m OM4 10G limit
- SFP+ SR transceivers for OM4 are significantly cheaper than LR transceivers for SM, reducing cost at the many short-distance switch-to-switch links inside buildings
- 850 nm VCSEL light sources used with OM4 are less expensive than the 1310 nm DFB lasers required for SM fibre

7.3 Cat6A UTP - Horizontal Cabling

- Rated for 10GBASE-T over the full 100 m channel length, future-proofing work area connectivity for the full 20–25 year design life
- Lower alien crosstalk (ANEXT) specification than Cat6 required for reliable 10G performance in bundled horizontal runs
- Terminated with 8P8C (RJ45) keystone jacks; universally compatible with all current and foreseeable end devices
- Current industry standard for new horizontal installations per TIA-568-C.2

7.4 Patch Panels

- Provide a physical management point: port-to-device mapping is documented at the panel; moves/adds/changes require only patch cord changes
- Labelled panels dramatically reduce fault diagnosis and resolution time
- Standard 1U rack form factor fits all TE rack cabinets

8. Features and Specifications of Routers and Switches

8.1 Cisco 3650-24PS - Layer 3 Distribution/Core Switch

Specification	Value
Ports	24 × 10/100/1000BASE-T (PoE+), 4 × 1G SFP (upgradeable to 10G)
Switching capacity	176 Gbps
Forwarding rate	65.5 Mpps
Layer 3 features	IPv4/IPv6 routing; OSPF, EIGRP, BGP, PBR
VLAN support	1024 VLANs
PoE budget	715 W total
Redundancy	Dual field-replaceable power supplies
Management	SNMPv2c/v3, NETCONF, REST API
Stacking	StackWise-480 (480 Gbps stacking bandwidth)

8.2 Cisco 2960-24TT - Layer 2 Access Switch

Specification	Value
Ports	24 × 10/100BASE-T + 2 × 10/100/1000BASE-T uplinks
Switching capacity	16 Gbps
Forwarding rate	6.5 Mpps
Layer	Layer 2 only
VLAN support	255 active VLANs
QoS	4 egress queues per port, 802.1p
Spanning Tree	PVST+, Rapid-PVST+, MSTP

8.3 Cisco 2911 ISR - Core Router

Specification	Value
WAN interfaces	3 × GE WAN (expandable with EHWIC modules)
Throughput	Up to 75 Mbps IPsec; 100 Mbps routing
Routing protocols	OSPF, BGP, EIGRP, RIPv2, IS-IS
Security	IOS Firewall, IPsec VPN, 802.1X NAC
Module support	EHWIC-1GE-SFP-CU for SM fibre backbone uplink

8.4 Cisco WLC-2504 - Wireless LAN Controller

Specification	Value
AP capacity	Up to 75 Lightweight APs
Client capacity	Up to 1000 wireless clients
Interfaces	4 × GE data + 1 × GE management
Security	WPA2-Enterprise (802.1X/EAP), RADIUS, rogue AP detection
Roaming	Layer 2 and Layer 3 fast roaming
Management	GUI, CLI, SNMP, Cisco Prime Infrastructure integration

9. Bill of Quantities

9.1 Backbone Network - Active Components

Table 4: Backbone Active Components

Item	Model	Qty	Notes
Backbone Distribution Switch	Cisco 3650-24PS	12	One per building node
Core Switch (CITeS)	Cisco 3650-24PS	2	Redundant pair at NOC
Border Router	Cisco 2911 ISR	2	Active/standby at CITeS
NGFW / Firewall	Cisco FPR-2110	2	HA pair at CITeS perimeter
Central WLC	Cisco WLC-2504	1	Manages all campus APs from CITeS
Campus-wide APs	Cisco 3702i	120	≈10 per building node avg
NMS / Core Servers	Dell PowerEdge R750	2	DNS, DHCP, RADIUS, NMS at CITeS

9.2 Backbone Network - Passive Components

Table 5: Backbone Passive Components

Item	Specification	Qty	Notes
OS2 SM Fibre Cable	9/125 μ m, outdoor armoured	10 000 m	Inter-building ring + redundant links
OM4 MM Fibre Cable	50/125 μ m, indoor	3 000 m	Building riser distribution

Item	Specification	Qty	Notes
HDPE Conduit	50 mm dia	10 000 m	Underground duct for backbone fibre
LC/UPC SM Patch Cords	Duplex, 3 m	120 pcs	SM transceiver connections at racks
LC/UPC MM Patch Cords	Duplex, 2 m	80 pcs	MM transceiver connections at racks
10G SFP+ LR Transceiver	SM, 10 km, 1310 nm	80 pcs	Inter-building SM links (×2 per link)
10G SFP+ SR Transceiver	MM, 300 m, 850 nm	60 pcs	Intra-building MM links
24-port LC Fibre Patch Panel	1U rack-mount	28 pcs	2 per backbone node + CITEs
Fibre Splice Closures	Outdoor-rated	30 pcs	Underground duct entry/exit points
42U Network Rack Cabinet	600×1000 mm	14 pcs	One per backbone node
1U Horizontal Cable Mgr.	Rack-mount	50 pcs	Cable dressing in racks
1 kVA Online UPS	Rack-mount	14 pcs	One per backbone node rack
Cable Ties / Labels	Standard	1 lot	Cable management consumables

9.3 ENTC Internal Network - Active Components

Table 6: ENTC Active Components

Item	Model	Qty	Notes
ENTC Core Switch	Cisco 3650-24PS	1	Layer 3 aggregation, inter-VLAN routing
Access Switch	Cisco 2960-24TT	12	One per floor zone / lab (Section 5)
Core Router	Cisco 2911 ISR	1	Backbone uplink gateway
Wireless LAN Ctrl.	Cisco WLC-2504	1	Manages 14 APs; administered from CITEs
Lightweight AP	Cisco 3702i	14	Per-floor deployment (Section 5)
DHCP/DNS Server	Server-PT (Svr0)	1	DHCP scopes for 3 VLANs; DNS relay

9.4 ENTC Internal Network - Passive Components

Table 7: ENTC Passive Components

Item	Specification	Qty	Notes
OM4 MM Fibre Cable	50/125 μ m, indoor	300 m	Vertical riser, Core Switch to floor TEs
Cat6A UTP Cable	23 AWG, 4-pair	2 500 m	Horizontal cabling to wall sockets
LC/UPC MM Fibre Patch Cords	Duplex, 2 m	40 pcs	Fibre patch panel to switch
RJ45 Cat6A Keystone Jacks	T568B wired	200 pcs	Wall outlet terminations
24-port Cat6A Patch Panel	1U rack-mount	10 pcs	Horizontal cabling termination at TEs
12-port LC Fibre Patch Panel	1U rack-mount	5 pcs	Fibre riser termination at MDF and TEs
10G SFP+ SR Transceiver	MM, OM4, 850 nm	30 pcs	Core Switch and floor switch uplinks
18U Wall-Mount Rack	450 mm depth	7 pcs	One per floor TE (Basement to 3.5F)
18U Floor-Standing Rack	600×800 mm	1 pc	Ground floor MDF (Core, Router, WLC)
600 VA Rack-Mount UPS	Online	3 pcs	MDF + critical floor TEs
Surface HDPE Trunking	40×25 mm	400 m	Horizontal cable routing on walls
1U Horizontal Cable Mgrs.	Rack-mount	20 pcs	Cable dressing in TE racks
Cable Ties / Labels	Standard	1 lot	Cable management consumables

10. Simulation Results

10.1 ENTC Network Simulation

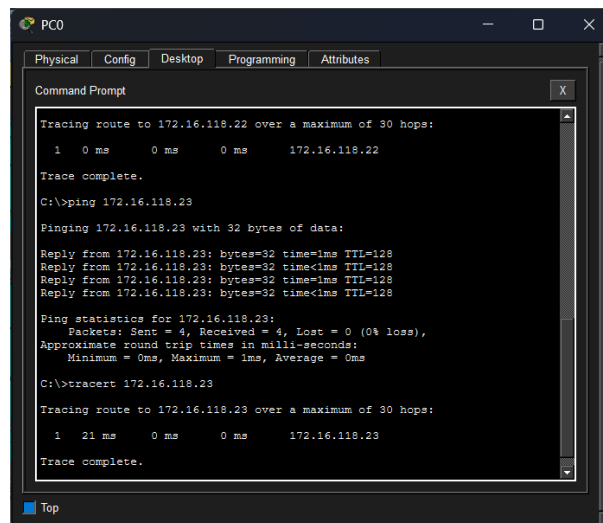
The ENTC network was implemented in Cisco Packet Tracer. Three VLANs (VLAN 10 Staff, VLAN 20 Student, VLAN 30 Labs) are configured on the Core Switch. 802.1Q trunks carry all three VLANs from the Core Switch to each floor access switch. DHCP scopes are configured on Server0 for all three VLAN subnets; Core Switch SVIs are DHCP relay agents. All 14 APs register to the WLC-2504 via CAPWAP. SSID-to-VLAN mappings are configured centrally on the WLC.

Connectivity Tests

Test 1 - Intra-VLAN

PC0 (VLAN 30, 172.16.118.21, Ground Floor) → PC1 (VLAN 30, 172.16.118.23, Lower Ground Floor).

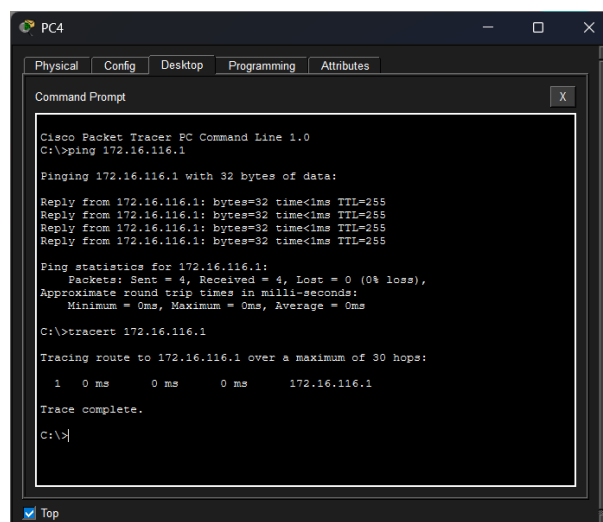
Result: Verifies Layer 2 switching within the same VLAN across floors.



Test 2 - Inter-VLAN

PC4 (VLAN 30 Labs, 1st Floor) → PC3 (VLAN 20 Staff, 2nd Floor). Packets routed by Core Switch SVI.

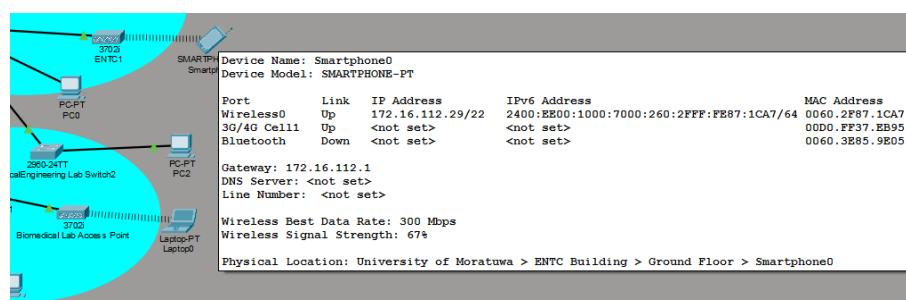
Result: Verifies inter-VLAN routing.



Test 3 - Wireless client association

Smartphone0 (Ground Floor) associates to ENTC1 AP on SSID student → VLAN 10; receives DHCP address in the 172.16.112.128/25 range.

Result: Correct VLAN assignment and DHCP address confirmed.



Test 4 - DHCP verification

All PCs receive correct subnet addresses and default gateway from Server0 via DHCP relay.

10.2 Backbone Network Simulation

The backbone ring is implemented in Cisco Packet Tracer as visible in Figure 2. OSPF Area 0 is configured on all 12 distribution switches and the 2 core switches at NOC. All building subnets are redistributed into OSPF.

*Connectivity Tests***Test 1 - Cross-building ping**

PC at ENTC (172.16.8.x) → PC at Sumanadasa Building (172.16.38.x). OSPF selects the shortest path. **Result:** 4/4 packets received; RTT avg 6–41 ms.

Test 2 - Ring failover

The direct NOC–ENTC link is administratively shut down. OSPF reconverges within 40 seconds, rerouting ENTC traffic via Architecture Department → NOC. **Result:** 4/4 packets received on alternate path after OSPF convergence.

11. Packet Tracer Simulation Files

The Cisco Packet Tracer simulation files can be downloaded from <https://dms.uom.lk/s/w73mTkCQKZENQ2y>.

These files contain all device configurations including IP addresses, VLAN definitions, 802.1Q trunks, DHCP server, and Wireless LAN Controller settings.

12. Conclusion

The designed backbone network provides a scalable, redundant, and cost-optimised network infrastructure for the University of Moratuwa for a 20–25 year design lifetime. The distributed ring topology with redundant shortcut links, and dual core switches at CITeS, ensures that no single link or node failure disrupts campus-wide connectivity. The choice of OS2 single-mode fibre for all inter-building links, combined with HDPE underground conduit, protects the physical investment for the full design life and allows seamless transceiver-only upgrades from 10 Gbps to 40 Gbps or 100 Gbps as bandwidth demands grow - without re-cabling the campus.

The ENTC internal LAN serves as the reference implementation. Its hierarchical two-tier design, functional three-VLAN segmentation (Staff/Student/Labs), centralised wireless management from CITeS via CAPWAP and the WLC-2504, and TIA-568-C structured cabling with OM4 fibre risers and Cat6A horizontal runs provide a template directly applicable to all other building nodes in the backbone.

Cisco Packet Tracer simulation confirms correct IPv4 and IPv6 end-to-end connectivity, inter-VLAN routing, DHCP, wireless client association and VLAN mapping, and OSPF ring failover with reconvergence within 40 seconds on link failure.

References

- [1] S. Perera, “Design of Networks,” lecture slides for EN2150 – Communication Network Engineering, adapted from A. Pasqual, Semester 4, Batch 23, Department of Electronic and Telecommunication Engineering, University of Moratuwa, available via Moodle (restricted access), 2026.
- [2] Cisco Systems, “Cisco Catalyst 3650 Series Switches,” Data Sheet, Cisco Systems. [Online]. Available: https://www.cisco.com/c/en/us/products/collateral/switches/catalyst-3650-series-switches/data_sheet-c78-729449.html [Accessed: Mar. 20, 2026].
- [3] Cisco Systems, “Cisco Catalyst 2960 Series Switches – Data Sheets,” Cisco Systems. [Online]. Available: <https://www.cisco.com/c/en/us/products/switches/catalyst-2960-series-switches/datasheet-listing.html> [Accessed: Mar. 20, 2026].
- [4] Cisco Systems, “Cisco 2500 Series Wireless Controllers Data Sheet,” Data Sheet, Cisco Systems. [Online]. Available: https://www.cisco.com/c/en/us/products/collateral/wireless/2500-series-wireless-controllers/data_sheet_c78-645111.html [Accessed: Mar. 20, 2026]. :contentReference[oaicite:0]index=0
- [5] “COMPLETE Campus Area Network System Design,” YouTube video. [Online]. Available: <https://www.youtube.com/watch?v=VrfhC9qpN64> [Accessed: Mar. 14, 2026].